

Ruffed Grouse Brood Habitat Use in Mixed Softwood–Hardwood Nordic–Temperate Forests, Quebec, Canada

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ABSTRACT Adequate cover is a critical component of ruffed grouse (*Bonasa umbellus*) habitat during the brood-rearing period when chick mortality is high. We assessed habitat use by ruffed grouse during the brood-rearing period by comparing characteristics of tree, shrub, and ground layers at ruffed grouse brood and random locations. We captured and radiomarked 29 females with broods in 2 forest settings of the Réserve faunique de Portneuf, Quebec, Canada. We described grouse habitat using ground surveys and forest maps, and we identified the used habitat characteristics using analysis of variance and logistic regression. Females with broods used mixed and regenerated clearcut stands that were 1.5–7 m tall and 11–20 years old. Compared with random locations, grouse locations had higher lateral obstruction (76% vs. 68%), higher small-stem density (29,085 stems/ha vs. 19,340 stems/ha), and were closer to roads and trails. Percentage of coverage by ground vegetation was not higher at grouse locations as often reported in previous studies. Results from this study will help orient ruffed grouse habitat management on Quebec public land and elsewhere in nordic–temperate mixed hardwood–softwood forests to maintain suitable brood habitat after logging operations. Forest management should promote growth of young mixed stands with high horizontal and vertical cover provided by high small-stem density, which offers protection against aerial and terrestrial predation. Edges such as roadsides are also important in brood habitat as they provide food and cover. (JOURNAL OF WILDLIFE MANAGEMENT 71(1):87–95; 2007)

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Availability of suitable brood-rearing habitat directly affects ruffed grouse (*Bonasa umbellus*) survival (Bump et al. 1947, Berner and Gysel 1969). Proximity and spatial configuration of essential resources is critical to ruffed grouse during brood rearing (McGarigal and McComb 1995, Schulte and Neimi 1998). For example, proximity of good quality habitats may decrease the energetic costs of food gathering and reduce predation risks (Kurzejeski and Root 1988, Thompson and Fritzell 1989, Small et al. 1993). Previous studies have described ruffed grouse brood habitat in Great Lakes states dominated by aspen (*Populus* spp.) forests (Bump et al. 1947, Godfrey 1975, Kubisiak 1978, Maxson 1978). These nonquantitative studies report the use of young forest stands with high stem density or older stands with a well-developed understory of shrubs and tree regeneration or ground cover. The research also identified forest openings and habitat edges as important components of brood habitat. Thompson et al. (1987), Scott et al. (1998), Haulton et al. (2003), and Tirpak et al. (2005) have published empirical studies of habitat use by ruffed grouse broods in the central and southern Appalachians in forests dominated by oak (*Quercus* spp.), hickory (*Carya* sp.), and aspen. These studies showed that ruffed grouse broods did not use habitat according to availability, suggesting they selected habitats. In northern regions such as Quebec, where

forests dominated by mixed hardwood–softwood stands in which commercial logging activities are widespread, researchers have not yet studied habitat use by hens with broods. Efficient management of grouse habitat in Quebec will require a better understanding of the species' northern habitat relationships, especially information concerning habitat features inside the activity area (third-order selection of Johnson 1980).

In Quebec, approximately 92% of land is of public ownership and over half contains commercial forests. During the past decades, extensive silvicultural practices such as large-scale clearcut logging, conifer planting, and precommercial thinning have substantially modified the suitability of landscape available to ruffed grouse, which depends on a juxtaposition of several habitat types in close proximity (Bump et al. 1947, Gullion and Svoboda 1972, Gullion 1977). In Quebec, consumptive recreation such as hunting and fishing do occur on public lands open to commercial forestry activities. However, because big-game opportunities have reached their full management potential, managers need to concentrate on methods and strategies to improve or enhance small-game hunting (e.g., ruffed grouse) to provide additional recreational user days in these systems. We suggest that management of public land should use a modified version of Gullion and Svoboda's (1972) classic mosaic of 1-ha clearcuts. They developed this management model for aspen forests, but it is questionable for use in modern forestry (Kubisiak et al. 1980, McCaffery et al. 1996) because it does not improve ruffed grouse habitat in hardwood–softwood forests, and it is not

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economically suitable for standard commercial logging in Quebec.

Our objective was to characterize habitat used by female ruffed grouse and their broods in 2 forest settings (study sites) of the Réserve faunique de Portneuf, Quebec, Canada. The 2 study sites differed mostly by stand age, height, and past forest interventions. We compared habitat use with availability in both sites to identify important brood habitat characteristics. We hypothesized that habitat use by female ruffed grouse was influenced by stand age, height, and past interventions found in study sites.

STUDY AREA

We conducted our study in 2 study sites, Belles-de-Jour (BDJ; 23 km²) and Landry (LAN; 17 km²), in the Réserve faunique de Portneuf, Quebec, Canada (47°11'N, 72°16'W). Three major vegetation zones were present in the study sites: sugar maple (*Acer saccharum*) and yellow birch (*Betula alleghaniensis*), balsam fir (*Abies balsamea*) and yellow birch, and balsam fir and white birch (*B. papyrifera*). Balsam fir and white birch stands were mostly present in LAN. Open areas, roads, trails, lakes, and streams were scattered throughout both study sites. Elevation ranged from 150 m to 650 m with gentle to steep slopes.

The 2 study sites differed mostly by stand age, height, and past forest interventions, as outlined by ground surveys of randomly distributed sample plots (Table 1). Landry was composed of a greater proportion of young stands than BDJ. The tree layer in both areas was composed mostly of balsam fir, yellow birch, black spruce (*Picea mariana*), sugar maple, red maple (*A. rubrum*), and white birch. Trembling and large-toothed aspen (*Populus tremuloides*, *P. grandidentata*) accounted for <1% of total basal area. The shrub layer in both sites was composed mostly of mountain maple (*A. spicatum*), mooseberry (*Viburnum alnifolium*), pin cherry (*Prunus pensylvanica*), beaked hazelnut (*Corylus cornuta*), ground hemlock (*Taxus canadensis*), berry species (*Rubus* spp.), and the previously mentioned tree species.

Grouse population density and hunting success were relatively low, as shown by the historical fall hunting success in the Réserve faunique de Portneuf. Hunting success was at its lowest in the last 9 years and spring counts of drumming males averaged 9 males/km² (3.6/40 ha) in both study sites (P. Blanchette, Ministère des Ressources naturelles et de la Faune du Québec, unpublished data).

METHODS

Capture and Telemetry

Searchers captured females on roads or trails during the spring and summer of 2001. We equally distributed searching effort across all roads and trails in both study sites. We first lured female grouse using a recorded lost chick distress call (Lyons 1981), and then we captured them with a fishing net or snare perch. We marked adult female grouse with 2 numbered plastic leg bands and fitted each bird with a R1-2BM Holohil transmitter (Holohil Systems Ltd., Carp, ON, Canada) using a backpack kevlar harness

Table 1. Forest stand characteristics of the study sites Belles-de-Jour (BDJ) and Landry (LAN) in the Réserve faunique de Portneuf, Quebec, Canada, 2001.

Forest characteristics	% random locations (no.)	
	BDJ (74)	LAN (73)
Stand age		
2–20 yr old	18	33
20–60 yr old	7	21
60–100 yr old	16	30
Uneven-aged stands	59	16
Stand ht		
1.5–7 m	23	40
7–18 m	28	37
>18 m	49	23
Commercial logging		
Uncut	47	47
Clearcuts 2–10 yr old	8	6
Clearcuts 11–20 yr old	10	28
Partial cuts (<20 yr old) ^a	35	19

^a Partial cuts: selection cutting, commercial thinning, improvement cutting, partial cutting, and shelterwood cutting.

(15–17 g; Brander 1968). We released birds at their capture site within 10 minutes of capture. The Quebec Faune et Parcs committee on animal care approved all bird handling techniques (certificate no. 00-01).

We located radiomarked grouse 2–3 times per week from 14 June to 10 September 2001. We considered mid-September as the onset of brood break-up and dispersal (Chambers and Sharp 1958, Godfrey and Marshall 1969). Grouse were located in a random order between 0630 hours and 1700 hours because ruffed grouse are inactive at night (Maxson 1977, Pietz and Tester 1982). We used a 3-element handheld Telonics antenna and an ATS R2100 (Advanced Telemetry Systems Inc, Isanti, MN) receiver to locate grouse. We took care not to scare off individuals to minimize disturbance and accurately locate hens with broods. We recorded geographical coordinates using a Garmin XL Global Positioning System (GPS; 15-m accuracy) and we attached an identified colored flag to the nearest tree for further habitat measurements.

Habitat Measurements

We took measurements at grouse locations 1 week after brood monitoring (between late Jun to late Aug 2001) to minimize disturbance, whereas we sampled random sites in July and August of the following year. At this northern latitude, one growing season probably does not change stand structure and composition. Using Arcview 3.2, we generated random locations in each study site (73 in BDJ and 74 in LAN) within the convex polygon encompassing all 2001 grouse locations. We rejected random locations falling on roads, water bodies, and inaccessible areas. We transferred geographical coordinates of random locations to a Garmin XL GPS for location in the field. We adapted habitat

Table 2. Habitat characteristics of ruffed grouse locations ($n = 355$) versus random locations ($n = 147$) in the Réserve faunique de Portneuf, Quebec, Canada, 2001.

Habitat variables	% use	% available	95% CI	Use vs. availability
Stand type				
Deciduous	35.8	40.1	-0.05 to 0.14	=
Mixed	58.0	40.8	-0.27 to -0.08	+
Coniferous	6.2	19.0	0.06 to 0.20	-
Stand age				
2–20 yr old	55.7	25.3	-0.39 to -0.21	+
20–60 yr old	9.7	13.7	-0.02 to 0.11	=
60–100 yr old	10.9	22.6	0.04 to 0.20	-
Young uneven-aged stands	15.8	19.9	-0.03 to 0.12	=
Old uneven-aged stands	8.8	17.8	0.02 to 0.16	-
Stand ht				
1.5–4 m	29.9	17.7	-0.20 to -0.04	+
5–7 m	24.8	13.6	-0.18 to -0.04	+
8–12 m	5.1	6.8	-0.03 to 0.06	=
13–17 m	15.5	25.9	0.02 to 0.18	-
≥18 m	24.8	36.1	0.02 to 0.20	-
Commercial logging				
Clearcuts 2–10 yr old	6.2	6.8	-0.04 to 0.05	=
Clearcuts 11–20 yr old	46.5	18.5	-0.36 to -0.20	+
Partial cuts	13.8	27.4	0.06 to 0.22	-
Uncut	33.5	47.3	0.04 to 0.23	-

^a (-), (+), and (=) indicate habitat categories used in less, greater, or equal proportion to availability, respectively.

measurements from Dunn and Braun (1986), Thompson et al. (1987), and Scott et al. (1998).

We recorded stand type, stand height, and presence of commercial logging (clearcuts or partial cuts) within a 0.04-ha circular plot centered on each grouse location. We considered selection cutting, commercial thinning, improvement cutting, partial cutting, and shelterwood cutting as partial cuts. We determined stand age, commercial logging year, distance to nearest clearcut, and distance to nearest road from recent digitized topographic forest maps (scale 1:20,000) produced by the Ministère des Ressources naturelles et de la Faune du Québec using ArcView Geographic Information System (GIS) 3.2 software.

We estimated deciduous, coniferous, and total canopy cover (>4 m tall), midstory cover (1–4 m tall), and lower-story cover (<1 m tall) using the point intercept method at 3-m intervals along a north–south 30-m transect centered on the location point (vertical cover). We assessed lateral obstruction using a 0.5 × 2.0-m density board divided into 4 0.5 × 0.5-m squares and placed 15 m from the standing observer (Nudds 1977). We assessed lateral obstruction in each cardinal direction. We estimated percentage of ground cover (<0.5 m tall) of leafy herbaceous plants, grasses, ferns, and cryptogams (moss and lichen) in 5 4-m² plots, positioned in each cardinal direction and at the center of the 0.04-ha plot. We included saplings and shrubs <0.5 m tall in total ground cover. We determined species richness of small stem species (≥50 cm tall and <9 cm dbh) and herbaceous species inside the 5 4-m² plots. We determined diameter at breast height and tree density (≥9 cm dbh) by total stem counts in the 0.04-ha circular plot. We measured

density of small woody species (>50 cm tall and <9 cm dbh), including shrubs, in each of the 5 4-m² plots.

Data Analysis

We derived 36 habitat variables from the habitat measurements. We pooled data by study site in order to consider specific site effects. We analyzed categorical variables using a chi-square test for homogeneity and simultaneous 95% confidence intervals (Marcum and Loftsgaarden 1980, Jelinski 1991). For each categorical variable, we used a log-linear model generalized for Poisson distribution to examine whether grouse use patterns differed between study sites (Heisey 1985). When log-linear models indicated no difference between study sites, we analyzed the variable by combining data from both sites.

We analyzed continuous variables using 2-way analysis of variance (ANOVA), and we used each variable's site and location type (grouse vs. random) as factors. When the study site factor affected the use-availability relationship, we made multiple comparisons separately for both sites using a least-square means test. Data were square root-transformed to tend toward normality and homogeneity of variances, but the large sample size makes the ANOVA robust to departure of normality.

Because we captured female ruffed grouse along roads, which were mostly located near clearcuts, we determined distance to forest roads using an additional set of random locations generated within each grouse home range. The number of random locations was equivalent to the number of grouse locations used to estimate home range size. We used a paired *t*-test to compare distance to forest roads in random locations and grouse locations.

We used a stepwise logistic regression analysis to generate

Table 3. Habitat characteristics of ruffed grouse locations versus random locations in the Réserve faunique de Portneuf, Quebec, Canada, 2001.

Habitat variables	Combined sites		Belles-de-Jour		Landry	
	Used (<i>n</i> = 355)	Random (<i>n</i> = 147)	Used (<i>n</i> = 170)	Random (<i>n</i> = 74)	Used (<i>n</i> = 185)	Random (<i>n</i> = 73)
	$\bar{x}$ SE	$\bar{x}$ SE	$\bar{x}$ SE	$\bar{x}$ SE	$\bar{x}$ SE	$\bar{x}$ SE
Cover (%)						
Total canopy closure (>4 m)			62.4	71.7	34.5	57.4**
			2.5	3.8	2.4	3.8
Deciduous canopy closure	37.3	47.2**				
	1.7	2.6				
Coniferous canopy closure			20.8	19.7	12.5	24.5**
			1.9	2.9	1.8	2.9
Total midstory cover (1–4 m)	71.1	69.8				
	1.2	1.9				
Mid deciduous cover	58.2	53.3				
	1.5	2.3				
Mid coniferous cover	20.1	26.4**				
	1.1	1.7				
Total lower story cover (<1 m)	38.4	52.1**				
	1.3	2.0				
Lower deciduous cover	22.3	35.5**				
	1.1	1.7				
Lower coniferous cover			15.1	13.8	13.2	20.7**
			1.3	2.0	1.2	2.0
Lateral obstruction (%; 0–2 m)	75.9	67.9**				
	1.0	1.5				
Herbaceous ground cover (%; <50 cm)						
Total ground cover	46.2	43.1				
	1.0	1.9				
Herbaceous plants	15.5	14.8				
	0.6	1.1				
Grasses	8.4	3.4				
	0.6	0.8				
Ferns	8.6	8.9				
	0.4	0.8				
Cryptograms (mosses & lichens)	17.9	18.8				
	0.8	2.0				
No. of herbaceous species	6.7	5.9				
	0.1	0.2				
Large stems (dbh ≥9 cm)						
Total basal area (m ² /ha)	8.2	14.9**				
	0.5	0.7				
Deciduous stems	4.7	8.6				
	0.3	0.3				
Coniferous stems			4.3	5.4	2.6	7.2**
			0.4	0.8	0.3	1.1
Total stem density (stems/ha)			342	498**	207	527**
			23	35	22	35
Deciduous stems	132	211				
	7	16				
Coniferous stems			162	219	119	383**
			14	31	13	59
Small stems (≥50 cm tall and dbh <9 cm)						
Total stem density (stems/ha)	29,085	19,340**				
	783	1,216				
Noncommercial stems	16,125	11,526**				
	785	1,220				
Shade-intolerant stems			294	405	5,070	2,205**
			360	546	345	550
Shade-tolerant stems			5,235	2,466**	2,319	2,043
			334	506	320	509
Coniferous stems	5,887	3,232**				
	294	456				

Table 3. Continued.

Habitat variables	Combined sites		Belles-de-Jour		Landry	
	Used (<i>n</i> = 355)	Random (<i>n</i> = 147)	Used (<i>n</i> = 170)	Random (<i>n</i> = 74)	Used (<i>n</i> = 185)	Random (<i>n</i> = 73)
	$\bar{x}$ SE	$\bar{x}$ SE	$\bar{x}$ SE	$\bar{x}$ SE	$\bar{x}$ SE	$\bar{x}$ SE
Dbh <1 cm (stems/ha)	21,300 696	14,160** 926				
<i>Rubus</i> spp. (stems/ha)	7.6 0.1	5.6** 0.2				
Distance to edge (m)						
Road			93 7	212** 11	27 7	154** 11
Clearcut 2–20 yr old	98 15	291** 23				

* $P < 0.05$. ** $P < 0.01$.

a ruffed grouse habitat model for both study areas combined. We retained all variables with $P < 0.15$ in previous univariate analyses for investigation (Hosmer and Lemeshow 2000), and we tested for multicollinearity using the condition index statistic ($r < 0.30$; Belsley et al. 1980). We removed the least significant variable in the correlated pairs from further analyses. We used the fractional polynomials method (Royston and Altman 1994) to determine whether previously selected explanatory variables were linearly related to the logit. When necessary, we transformed variables using the best transformation suggested by the fractional polynomials method. We included variables exhibiting a normal or skewed distribution as a linear term in the model and as a transformed term of the variable.

Logistical problems prevented the use of independent locations to test our model. We therefore calculated the percentage of locations classified correctly (relocated vs. random) by the logistic regression model using the leave-one-out cross-validation method (Stone 1977). This pseudovalidation procedure classified each observation using the logistic function developed from all observations, except the one being classified. We repeated this procedure for each of the 496 observations. We considered locations with a predictive probability of ≥ 0.70 (no. grouse occurrences/total no. observations) to predict grouse presence. We performed statistical analyses using SAS/STAT 8.02 (SAS Institute 1999). We considered results significant when $P < 0.05$.

RESULTS

We followed 29 hens with broods; 14 in BDJ and 15 in LAN. We captured most females (26) between 12 June and 3 July 2001, but we captured 3 females between 25 July and 10 August 2001 to replace lost or dead individuals. We determined habitat use using 12.2 ± 5.8 ($\bar{x} \pm \text{SD}$) relocations per grouse. We performed analyses using 355 independent grouse locations (170 BDJ, 185 LAN) and 147 random locations.

Hens with broods used mixed stands and avoided coniferous stands ($\chi^2 = 23.6$, $\text{df} = 2$, $P < 0.0001$; Table 2). Grouse used stands 2–20 years old and avoided stands 60–100 years old, as well as old, uneven-aged stands ($\chi^2 =$

40.0, $\text{df} = 4$, $P < 0.0001$). They used stands 1.5–7 m tall and avoided stands ≥ 13 m tall ($\chi^2 = 23.2$, $\text{df} = 4$, $P < 0.0001$). They used clearcuts 11–20 years old, whereas they avoided uncut mature stands (> 20 yr old) and partially cut stands ($\chi^2 = 37.2$, $\text{df} = 3$, $P < 0.0001$; Table 2).

Canopy cover (> 4 m tall) at grouse locations was 62% in BDJ and 35% in LAN (Table 3). In LAN, total canopy cover at grouse locations was lower than at random locations. Total percentage of midstory cover (1–4 m tall) was 71% for both sites combined and did not differ from random locations. Total percentage of lower-story cover (< 1 m tall) at both sites was lower than at random locations (Table 3). Deciduous trees dominated at grouse locations, and coniferous trees occupied $< 21\%$ of cover. Lateral obstruction was greater at grouse locations than at random locations (Table 3). Habitat use by broods was not related to herbaceous ground cover. Species richness of small stem species and herbaceous species was greater at grouse locations than at random locations (Table 3).

Total tree density and basal area ($\text{dbh} \geq 9$ cm) were lower at grouse locations than at random locations (Table 3). Coniferous diameter at breast height and stem density were lower at grouse locations than at random locations in LAN (Table 3). Density of small woody stems ($\text{ht} \geq 50$ cm and $\text{dbh} < 9$ cm), especially those < 1 cm diameter at breast height, was greater at grouse locations (29,085 stems/ha) than at random locations (19,340 stems/ha). Density of noncommercial small woody species and berry species (*Rubus* spp.) was greater at grouse locations than at random locations (Table 3). Grouse locations (54 ± 3 m) were closer to roads than random locations (67 ± 3 m; $t = -2.648$, $\text{df} = 16$, $P = 0.018$).

Stepwise logistic regression analysis produced a model using 8 variables to predict the presence of hens with broods (receiving operating characteristic curve = 0.93; Homer-Lemeshow statistics $\chi^2 = 8.84$, $\text{df} = 8$, $P = 0.356$; Table 4). The probability of locating ruffed grouse broods was greater near roads and clearcuts 11–20 years old, as well as in stands with more small woody species and less midstory coniferous cover (1–4 m tall). A lower-story deciduous cover of 0–30% positively influenced grouse presence, whereas a cover of 30–

Table 4. Results of the stepwise logistic regression model predicting ruffed grouse locations versus random locations in the Réserve faunique de Portneuf, Quebec, Canada, 2001.^a

Variable (transformation)	Estimate	SE	P value	Wald χ^2
Intercept	3.5550	0.7290	<0.0001	
Distance to road (square-root)	-0.2711	0.0363	<0.0001	55.899
Lower deciduous cover <1 m tall	-2.0942	0.2858	<0.0001	53.679
No. of small woody species	0.5359	0.0800	<0.0001	44.890
Lower deciduous cover <1 m tall (log)	1.5016	0.3065	<0.0001	24.009
Distance to clearcut 11–20 yr old (log)	-0.2089	0.0428	<0.0001	23.814
Mid coniferous cover 1–4 m tall (x^3)	-0.0416	0.0086	<0.0001	23.648
Deciduous canopy cover >4 m tall	0.9620	0.3271	0.0033	8.652
Deciduous canopy cover >4 m tall (x^2)	-0.1394	0.0618	0.0242	5.083

^a Receiving operating characteristic curve = 0.93; Homer–Lemeshow statistics $\chi^2 = 8.84$, df = 8, $P = 0.356$.

100% reduced probability of finding ruffed grouse broods. A deciduous canopy cover of 0–40% positively influenced grouse presence, whereas a cover of 40–100% reduced the probability of encountering grouse broods. Leave-one-out cross-validation method pseudovalidation procedures resulted in the correct classification of 84.7% of grouse locations and 81.7% of random locations.

DISCUSSION

In our study, ruffed grouse broods used mixed dense clearcut stands (11–20 yr old) of 1.5-m to 7-m height. Moreover, in one study site, ruffed grouse broods also used uneven-aged stands with high percentage of overstory and midstory covers and higher coniferous cover and tree stem density than the other site. These differences may be attributed to lower availability of young stands in BDJ, resulting in the use of alternative habitats such as uneven-aged stands and coniferous cover, which likely provided suitable protection from predators. Past forest interventions determined availability of young stands in a site and thus influenced habitat use by ruffed grouse broods.

Researchers have studied ruffed grouse brood-rearing habitat in disparate geographic regions or habitat types (Table 5). Although some habitat characteristics differed among these studies, many commonalities are present. Gullion (1977) suggested that stands <10 years old were the optimal brood habitat. In the Portneuf area, colder temperatures shorten the growing season, which may partially explain the use of clearcuts 11–20 years old instead of younger clearcuts as observed in United States (Dessecker and McAuley 2001). Similarly to other studies (Bump et al. 1947, Gullion 1977, Kubisiak 1978, Scott et al. 1998), grouse locations in this study had higher sapling and shrub densities than random locations (Table 5). In Minnesota (aspen stands), Pennsylvania (oak stands), and Wisconsin, USA (oak and aspen stands), broods consistently used habitat with a high density of small stems or shrubs. On the other hand, in Idaho, USA (aspen–conifer stands) and in Missouri, USA (oak stands), broods used habitat with a lower density of small stems or shrubs than in our study sites (Stauffer and Peterson 1985, Thompson et al. 1987). These authors suggest that stem density was lower than in other studies because few early successional stands were present in their study area. In Virginia, USA, broods used oak stands

with comparable stem density (Haulton et al. 2003). In their study, broods did not select stands with higher stem density although recently cut stands (<20 yr old) with high stem density were available to broods. Authors suggest that habitat used may have been characteristic of brood use on the southern periphery of the ruffed grouse geographic range (Haulton et al. 2003).

Previous studies have found ground cover to be an important component of brood habitat (Berner and Gysel 1969, Stauffer and Peterson 1985, Thompson et al. 1987, Doyon 1992, Haulton et al. 2003; Table 5). Indeed, female ruffed grouse and their chicks feed and travel primarily on the ground (Bump et al. 1947), and ground cover has been shown to increase insect availability or protection from predators (Stewart 1956, Thompson et al. 1987, Haulton et al. 2003). Researchers have found rich herbaceous vegetation to increase arthropod availability (Healy 1985, Hollifield and Dimmick 1995). Haulton et al. (2003) reported that broods used sites with greater arthropod abundance during the first 3 weeks posthatch than sites used during the following weeks. In habitats where cover of small woody stems was low, grouse may have relied on high ground cover and the canopy's horizontal cover for protection and food (Thompson et al. 1987). Stauffer and Peterson (1985), Thompson et al. (1987), Doyon (1992), and Haulton et al. (2003) also reported the use of habitat with high ground cover in the absence of high stem density provided by suitable-age regeneration. In our study, broods used sites with low ground cover of herbaceous plants, ferns, or grasses (8–32% cover), as well as low woody cover <1 m tall (38%). Our results were comparable to Scott et al. (1998) in stands with high stem density. Porath and Vohs (1972) and Kubisiak (1978) have observed the avoidance of extremely dense ground cover by grouse. Gullion (1977) suggested that ground cover must not be too dense to maintain effective surveillance of the terrain by females and to reduce terrestrial predation. This may explain why grouse avoided stands with deciduous cover (<1 m tall) >30%.

Vegetative cover, small-stem species richness, and the presence of habitat edges and openings were the most important characteristics of ruffed grouse brood habitat and accounted for 84.7% of grouse locations correctly classified by logistic regression. In previous studies, researchers did not find that tree canopy cover influenced brood-rearing

Table 5. Ruffed grouse brood habitat characteristics according to 9 North American studies.

Study site	Stand type	Canopy cover		Ground cover		Stem density		Source
		%	Class	%	Class	Stems/ha	Class ^a	
Great Lakes								
MI	Aspen – hardwood			48	20–91 cm tall			Berner and Gysel 1969
MN	Aspen – hardwood					19,000–25,000	a, b	Gullion 1977
WI	Aspen; oak					39,000–83,000	a, b, c	Kubisiak 1978
Western								
ID	Aspen; aspen – conifer	61; 63		83; 70		4,005; 7,979	a, b	Stauffer and Peterson 1985
Central Appalachians								
PA	Oak;	78	≥2 m tall			21,400	a, b	Scott et al. 1998
	aspen – oak	78	<2 m tall	25				
PA	Pin cherry - black birch – red maple – aspen	81	Dbh >8 cm	69	<1 m tall	18,671	a <8 cm dbh	Tirpak et al. 2005
Southern Appalachians								
MO	Oak–hickory	80		65		5,797	a, c	Thompson et al. 1987
VA ^b	Oak–chestnut	79	≥2 m tall	60	<0.5 m tall	5,435	a, c	Haulton et al. 2003
Northern								
Île aux Lièvres,	Balsam fir–white birch	69	≥2 m tall	64				Doyon 1992
PQ		36	<2 m tall					
Portneuf, PQ ^c	Balsam fir–yellow birch	62 BDJ	>4 m tall	8–32	<0.5 m tall	29,427 BDJ	a, c	Present study
		35 LAN				29,292 LAN		
		71	1–4 m tall					
		38	<1 m tall					

^a a = trees; b = saplings; c = shrubs.^b Combined data for 3 successive 3-week periods.^c BDJ = Belles-de-Jour; LAN = Landry.

habitat selection by ruffed grouse (Stauffer and Peterson 1985, Thompson et al. 1987, Doyon 1992, Scott et al. 1998, Haulton et al. 2003). These studies reported canopy cover ranging from 61% to 80%, including both midstory and overstory trees (Table 5). Even though broods in our study used comparable canopy cover, our results suggest that cover provided by midstory trees (1–4 m tall) was most important to grouse (71%). Grouse consistently used high midstory cover in the 2 study sites, despite important variations in overstory canopy cover. Our study shows that broods use sites with a mid, lower, or ground dense cover, be it shrub or herbaceous. This cover offers protection from aerial predators and provides insects, the main food source for ruffed grouse chicks. The presence of a dense overstory (>4 m in this study) does not seem compatible with the presence of a dense shrub or herbaceous cover. The logistic regression analyses support this finding, which showed that deciduous overstory canopy closure >40% was less suitable for broods.

Many studies have reported forest openings and habitat edges to be important components of brood habitat (Berner and Gysel 1969, Porath and Vohs 1972, Maxson 1978, Scott et al. 1998, Fearer and Stauffer 2003). Secondary roads, trails, natural forest openings, and small clearings offer luxuriant growth of ground vegetation along their margins (Matlack 1994). High food (plant and insect) availability and adequate protection from predators may explain the use of these edges (Stewart 1956). In areas where aspen is absent or scarce, regenerated cuts may not supply

adequate food (Fearer 1999 in Fearer and Stauffer 2003); therefore, grouse may rely on adjacent edges bordering recent clearcuts (Fearer and Stauffer 2003). In our study, grouse locations were significantly closer to roads and forest cuts 2–20 years old than random locations. Sapling, shrub, and herbaceous plant species richness were also greater at grouse locations than at random locations. However, our data did not allow us to compare species richness under different habitat conditions (edges vs. actual regenerating stand), nor did our observations allow us to clearly identify edges as important brood feeding areas. We rarely saw broods directly in openings (fields, recent clearcuts, roads), as observed by Porath and Vohs (1972), Maxson (1978), and Scott et al. (1998).

MANAGEMENT IMPLICATIONS

To create brood-rearing habitat, we suggest using appropriate management techniques that promote growth of young deciduous or mixed even-aged stands with high stem density, such as clearcuts, seed tree cuts, or shelterwood cuts. Management techniques in uneven-aged forests should provide maximal canopy opening to promote a dense shrub layer. Clearcut patches should maximize the edge effect while maintaining sufficient cover area within the stand. Managers should avoid post-clearcut treatments, such as precommercial thinning, that dramatically reduce stem density and promote more valuable young coniferous stems to maintain the quality of brood-rearing habitat and to

maintain habitat for other species that depend on young and dense stands to survive (Bélanger 2000, Etcheverry et al. 2005).

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